Layout-Preserving Translation of PDF Documents inside a Code Editor

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Abstract. We describe a viewer that shows a PDF document and its translation side by side without changing the design of the page. The page is drawn twice by the same rendering engine, once as it is and once without text, and the translated text is placed over the second drawing. Figures, rules, and coloured backgrounds therefore stay exactly as they are in the source. The translation is produced block by block, so the reader sees the first paragraphs after a few seconds.

1 Introduction

Reading technical documents in a second language is slow, and switching between a viewer and a translation tool breaks concentration. Many readers therefore want the translation to appear next to the original, in the same position on the page, so that figures, tables, and equations can still be found at a glance.[1]

Existing tools usually follow one of two routes. The first converts the document to plain text or HTML and translates the result; this is simple, but the original design is lost. The second rebuilds a new PDF file with translated text; this keeps the design, but it needs a heavy pipeline, and the reader must wait until the whole file is finished [1, 2].

In this paper we take a third route. The page is drawn twice by the same rendering engine: once as it is, and once without any text. The translated text is then placed over the second image as ordinary web content. Because everything except the text comes from the original drawing commands, backgrounds, rules, and figures are identical to the source.

Our contributions are as follows:

- a rendering scheme that removes text without touching vector graphics or images;
- a simple geometric method for grouping text into paragraphs;
- a fitting procedure that keeps each translated paragraph inside its original box.

2 Method

2.1 Text-free rendering

A PDF page is a list of drawing operators. Let $P = (o_1, \dots, o_n)$ be this list and let $T \subset P$ be the operators that show text. We render the full list to obtain an image I_{full} and the filtered list $P \setminus T$ to obtain I_{bare}. For a translated region R the displayed image is

$$I(x) = \begin{cases} I_{\text{bare}}(x) & x \in R, \\ I_{\text{full}}(x) & \text{otherwise.} \end{cases} \tag{1}$$

Regions that are not translated, such as equations and labels inside figures, keep their original pixels.

2.2 Block segmentation

Text fragments are first merged into lines by comparing baselines, and lines are then merged into blocks when they share a font size, overlap horizontally, and follow the regular line pitch of the page. A new block starts at an indented line, at a list marker, or after a vertical gap.

2.3 Fitting the translation

Japanese text often needs slightly more room than English. For each block we search for the largest scale $k \in [0.7, 1]$ such that the translated paragraph does not exceed the height of the original box. If no such scale exists, the paragraph may extend into the empty space below it.

Note. The source file is never modified. Closing the translated view returns the reader to the original document.

3 Experiments

We measured the method on 120 pages taken from manuals, reports, and research papers. Table 1 shows the share of blocks that fitted into their original boxes, and Figure 1 shows how the time to the first translated block changes with the length of the page.

The share is lowest for research papers, because their pages are dense and leave little empty space below each paragraph.

[1] This is a synthetic sample document made for testing. All numbers in it are invented.

Table 1: Share of translated blocks that fit into the original box.

Document type	Pages	Blocks	Fit (%)
Manuals	40	1,120	96.4
Reports	40	980	93.1
Research papers	40	1,460	89.7

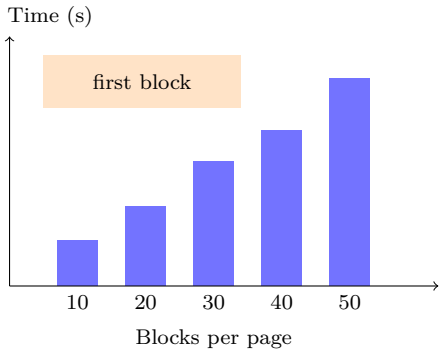


Figure 1: Time until the first translated block appears.

4 Limitations

Scanned pages contain no text operators, so they need optical character recognition before translation. Tables are translated cell by cell, which can break sentences that span several cells. Inline mathematics is sent to the model as plain text and may be altered.

5 Conclusion

Drawing the page twice and overlaying translated text keeps the design of the original document while remaining light enough to run inside an editor. We plan to add support for exporting the result as a new PDF file.

References

[1] A. Example and S. Sample. Rebuilding translated documents from an intermediate representation. *Journal of Document Engineering*, 12(3):45–67, 2024.

[2] C. Placeholder. A survey of layout analysis for born-digital documents. In *Proceedings of the Workshop on Document Layout*, pages 1–10, 2025.